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Title 21 — Food and Drugs

Chapter I — Food and Drug Administration, Department of Health and Human Services

Subchapter H — Medical Devices

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PART 892—RADIOLOGY DEVICES

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Subpart A—General Provisions

§ 892.1 Scope.

- (a) This part sets forth the classification of radiology devices intended for human use that are in commercial distribution.
- (b) The identification of a device in a regulation in this part is not a precise description of every device that is, or will be, subject to the regulation. A manufacturer who submits a premarket notification submission for a device under part 807 cannot show merely that the device is accurately described by the section title and identification provision of a regulation in this part but shall state why the device is substantially equivalent to other devices, as required by § 807.87.
- (c) To avoid duplicative listings, a radiology device that has two or more types of uses (e.g., use both as a diagnostic device and a therapeutic device) is listed in one subpart only.
- (d) References in this part to regulatory sections of the Code of Federal Regulations are to chapter I of this title 21, unless otherwise noted.
- (e) Guidance documents referenced in this part are available on the Internet at <http://www.fda.gov/MedicalDevices/DeviceRegulationandGuidance/GuidanceDocuments/default.htm>.

[53 FR 1567, Jan. 20, 1988, as amended at 73 FR 40969, July 17, 2008; 78 FR 18233, Mar. 26, 2013]

§ 892.3 Effective dates of requirement for premarket approval.

A device included in this part that is classified into class III (premarket approval) shall not be commercially distributed after the date shown in the regulation classifying the device unless the manufacturer has an approval under section 515 of the act (unless an exemption has been granted under section 520(g)(2) of the act). An approval under section 515 of the act consists of FDA's issuance of an order approving an application for premarket approval (PMA) for the device or declaring completed a product development protocol (PDP) for the device.

- (a) Before FDA requires that a device commercially distributed before the enactment date of the amendments, or a device that has been found substantially equivalent to such a device, has an approval under section 515 of the act, FDA must promulgate a regulation under section 515(b) of the act requiring such approval, except as provided in paragraph (b) of this section. Such a regulation under section 515(b) of the act shall not be effective during the grace period ending on the 90th day after its promulgation or on the last day of the 30th full calendar month after the regulation that classifies the device into class III is effective, whichever is later. See section 501(f)(2)(B) of the act. Accordingly, unless an effective date of the requirement for premarket approval is shown in the regulation for a device classified into class III in this part, the device may be commercially distributed without FDA's issuance of an order approving a PMA or declaring completed a PDP for the device. If FDA promulgates a regulation under section 515(b) of the act requiring premarket approval for a device, section 501(f)(1)(A) of the act applies to the device.
- (b) Any new, not substantially equivalent, device introduced into commercial distribution on or after May 28, 1976, including a device formerly marketed that has been substantially altered, is classified by statute (section 513(f) of the act) into class III without any grace period and FDA must have issued an order approving a PMA or declaring completed a PDP for the device before the device is commercially distributed unless it is reclassified. If FDA knows that a device being commercially distributed may be a "new" device as defined in this section because of any new intended use or other reasons, FDA may codify the statutory classification of the device into class III for such new use. Accordingly, the regulation for such a class III device states that as of the enactment date of the amendments, May 28, 1976, the device must have an approval under section 515 of the act before commercial distribution.

§ 892.9 Limitations of exemptions from section 510(k) of the Federal Food, Drug, and Cosmetic Act (the act).

The exemption from the requirement of premarket notification (section 510(k) of the act) for a generic type of class I or II device is only to the extent that the device has existing or reasonably foreseeable characteristics of commercially distributed devices within that generic type or, in the case of in vitro diagnostic devices, only to the extent that misdiagnosis as a result of using the device would not be associated with high morbidity or mortality. Accordingly, manufacturers of any commercially distributed class I or II device for which FDA has granted an exemption from the requirement of premarket notification must still submit a premarket notification to FDA before introducing or delivering for introduction into interstate commerce for commercial distribution the device when:

- (a) The device is intended for a use different from the intended use of a legally marketed device in that generic type of device; e.g., the device is intended for a different medical purpose, or the device is intended for lay use where the former intended use was by health care professionals only;
- (b) The modified device operates using a different fundamental scientific technology than a legally marketed device in that generic type of device; e.g., a surgical instrument cuts tissue with a laser beam rather than with a sharpened metal blade, or an in vitro diagnostic device detects or identifies infectious agents by using deoxyribonucleic acid (DNA) probe or nucleic acid hybridization technology rather than culture or immunoassay technology; or
- (c) The device is an in vitro device that is intended:
 - (1) For use in the diagnosis, monitoring, or screening of neoplastic diseases with the exception of immunohistochemical devices;
 - (2) For use in screening or diagnosis of familial or acquired genetic disorders, including inborn errors of metabolism;
 - (3) For measuring an analyte that serves as a surrogate marker for screening, diagnosis, or monitoring life-threatening diseases such as acquired immune deficiency syndrome (AIDS), chronic or active hepatitis, tuberculosis, or myocardial infarction or to monitor therapy;
 - (4) For assessing the risk of cardiovascular diseases;
 - (5) For use in diabetes management;
 - (6) For identifying or inferring the identity of a microorganism directly from clinical material;
 - (7) For detection of antibodies to microorganisms other than immunoglobulin G (IgG) or IgG assays when the results are not qualitative, or are used to determine immunity, or the assay is intended for use in matrices other than serum or plasma;
 - (8) For noninvasive testing as defined in § 812.3(k) of this chapter; and
 - (9) For near patient testing (point of care).

[65 FR 2322, Jan. 14, 2000]

Subpart B—Diagnostic Devices

§ 892.1000 Magnetic resonance diagnostic device.

- (a) **Identification.** A magnetic resonance diagnostic device is intended for general diagnostic use to present images which reflect the spatial distribution and/or magnetic resonance spectra which reflect frequency and distribution of nuclei exhibiting nuclear magnetic resonance. Other physical parameters derived from the images and/or spectra may also be produced. The device includes hydrogen-1 (proton) imaging, sodium-23 imaging, hydrogen-1 spectroscopy, phosphorus-31 spectroscopy, and chemical shift imaging (preserving simultaneous frequency and spatial information).
- (b) **Classification.** Class II (special controls). A magnetic resonance imaging disposable kit intended for use with a magnetic resonance diagnostic device only is exempt from the premarket notification procedures in subpart E of part 807 of this chapter subject to the limitations in § 892.9.

[53 FR 5078, Feb. 1, 1989, as amended at 84 FR 71818, Dec. 30, 2019]

§ 892.1001 Liver iron concentration imaging companion diagnostic for deferasirox.

- (a) **Identification.** The liver iron concentration imaging companion diagnostic for deferasirox is an image processing device intended to aid in the identification and monitoring of non-transfusion-dependent thalassemia patients receiving therapy with deferasirox. The device calculates a numeric value for liver iron concentration based on magnetic resonance images acquired under controlled conditions. The calculated numeric value is used to assess the need for deferasirox treatment and for monitoring treatment in patients with non-transfusion-dependent thalassemia. The liver iron concentration imaging companion diagnostic for deferasirox is essential to the safe and effective use of deferasirox in patients with non-transfusion-dependent thalassemia.
- (b) **Classification.** Class II (special controls). The special controls for this device are:
 - (1) Design verification and validation must include nonclinical and clinical performance testing demonstrating the bias, precision, repeatability, and reproducibility of liver iron concentration measurements.
 - (2) Labeling must include specifying:
 - (i) Instructions for acceptance testing of images prior to processing;
 - (ii) Data processing quality assurance protocols; and
 - (iii) The sensitivity and specificity of liver iron concentration measurements.

[90 FR 40735, Aug. 21, 2025]

§ 892.1100 Scintillation (gamma) camera.

- (a) **Identification.** A scintillation (gamma) camera is a device intended to image the distribution of radionuclides in the body by means of a photon radiation detector. This generic type of device may include signal analysis and display equipment, patient and equipment supports, radionuclide anatomical markers, component parts, and accessories.
- (b) **Classification.** Class I (general controls).

[55 FR 48443, Nov. 20, 1990, as amended at 66 FR 46953, Sept. 10, 2001]

§ 892.1110 Positron camera.

- (a) **Identification.** A positron camera is a device intended to image the distribution of positron-emitting radionuclides in the body. This generic type of device may include signal analysis and display equipment, patient and equipment supports, radionuclide anatomical markers, component parts, and accessories.
- (b) **Classification.** Class I (general controls).

[55 FR 48444, Nov. 20, 1990, as amended at 66 FR 46953, Sept. 10, 2001]

§ 892.1130 Nuclear whole body counter.

- (a) **Identification.** A nuclear whole body counter is a device intended to measure the amount of radionuclides in the entire body. This generic type of device may include signal analysis and display equipment, patient and equipment supports, component parts, and accessories.
- (b) **Classification.** Class I (general controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter, subject to the limitations in § 892.9.

[53 FR 1567, Jan. 20, 1988, as amended at 59 FR 63015, Dec. 7, 1994; 66 FR 38818, July 25, 2001]

[55 FR 48444, Nov. 20, 1990]

§ 892.1170 Bone densitometer.

- (a) **Identification.** A bone densitometer is a device intended for medical purposes to measure bone density and mineral content by x-ray or gamma ray transmission measurements through the bone and adjacent tissues. This generic type of device may include signal analysis and display equipment, patient and equipment supports, component parts, and accessories.
- (b) **Classification.** Class II.

§ 892.1180 Bone sonometer.

- (a) **Identification.** A bone sonometer is a device that transmits ultrasound energy into the human body to measure acoustic properties of bone that indicate overall bone health and fracture risk. The primary components of the device are a voltage generator, a transmitting transducer, a receiving transducer, and hardware and software for reception and processing of the received ultrasonic signal.
- (b) **Classification.** Class II (special controls). The special control for this device is FDA's "Guidance for Industry and FDA Staff; Class II Special Controls Guidance Document: Bone Sonometers." See § 892.1(e) for the availability of this guidance document.

[73 FR 40969, July 17, 2008]

§ 892.1200 Emission computed tomography system.

- (a) **Identification.** An emission computed tomography system is a device intended to detect the location and distribution of gamma ray- and positron-emitting radionuclides in the body and produce cross-sectional images through computer reconstruction of the data. This generic type of device may include signal analysis and display equipment, patient and equipment supports, radionuclide anatomical markers, component parts, and accessories.

- (b) *Classification.* Class II.

§ 892.1220 Fluorescent scanner.

- (a) *Identification.* A fluorescent scanner is a device intended to measure the induced fluorescent radiation in the body by exposing the body to certain x-rays or low-energy gamma rays. This generic type of device may include signal analysis and display equipment, patient and equipment supports, component parts and accessories.
- (b) *Classification.* Class II.

§ 892.1300 Nuclear rectilinear scanner.

- (a) *Identification.* A nuclear rectilinear scanner is a device intended to image the distribution of radionuclides in the body by means of a detector (or detectors) whose position moves in two directions with respect to the patient. This generic type of device may include signal analysis and display equipment, patient and equipment supports, radionuclide anatomical markers, component parts, and accessories.
- (b) *Classification.* Class I (general controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter, subject to the limitations in § 892.9.

[55 FR 48444, Nov. 20, 1990, as amended at 65 FR 2322, Jan. 14, 2000; 66 FR 38818, July 25, 2001]

§ 892.1310 Nuclear tomography system.

- (a) *Identification.* A nuclear tomography system is a device intended to detect nuclear radiation in the body and produce images of a specific cross-sectional plane of the body by blurring or eliminating detail from other planes. This generic type of devices may include signal analysis and display equipment, patient and equipment supports, radionuclide anatomical markers, component parts, and accessories.
- (b) *Classification.* Class II.

§ 892.1320 Nuclear uptake probe.

- (a) *Identification.* A nuclear uptake probe is a device intended to measure the amount of radionuclide taken up by a particular organ or body region. This generic type of device may include a single or multiple detector probe, signal analysis and display equipment, patient and equipment supports, component parts, and accessories.
- (b) *Classification.* Class I (general controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter subject to § 892.9.

[55 FR 48444, Nov. 20, 1990, as amended at 65 FR 2322, Jan. 14, 2000]

§ 892.1330 Nuclear whole body scanner.

- (a) *Identification.* A nuclear whole body scanner is a device intended to measure and image the distribution of radionuclides in the body by means of a wide-aperture detector whose position moves in one direction with respect to the patient. This generic type of device may include signal analysis and display equipment, patient and equipment supports, radionuclide anatomical markers, component parts, and accessories.
- (b) *Classification.* Class I (general controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter subject to § 892.9.

[55 FR 48444, Nov. 20, 1990, as amended at 65 FR 2322, Jan. 14, 2000]

§ 892.1350 Nuclear scanning bed.

- (a) **Identification.** A nuclear scanning bed is an adjustable bed intended to support a patient during a nuclear medicine procedure.
- (b) **Classification.** Class I (general controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter subject to § 892.9.

[55 FR 48444, Nov. 20, 1990, as amended at 59 FR 63015, Dec. 7, 1994; 65 FR 2322, Jan. 14, 2000]

§ 892.1360 Radionuclide dose calibrator.

- (a) **Identification.** A radionuclide dose calibrator is a radiation detection device intended to assay radionuclides before their administration to patients.
- (b) **Classification.** Class II.

§ 892.1370 Nuclear anthropomorphic phantom.

- (a) **Identification.** A nuclear anthropomorphic phantom is a human tissue facsimile that contains a radioactive source or a cavity in which a radioactive sample can be inserted. It is intended to calibrate nuclear uptake probes or other medical instruments.
- (b) **Classification.** Class I (general controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter, subject to the limitations in § 892.9.

[53 FR 1567, Jan. 20, 1988, as amended at 54 FR 13832, Apr. 5, 1989; 66 FR 38818, July 25, 2001]

§ 892.1380 Nuclear flood source phantom.

- (a) **Identification.** A nuclear flood source phantom is a device that consists of a radiolucent container filled with a uniformly distributed solution of a desired radionuclide. It is intended to calibrate a medical gamma camera-collimator system for uniformity of response.
- (b) **Classification.** Class I (general controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter, subject to the limitations in § 892.9.

[53 FR 1567, Jan. 20, 1988, as amended at 54 FR 13832, Apr. 5, 1989; 66 FR 38819, July 25, 2001]

§ 892.1390 Radionuclide rebreathing system.

- (a) **Identification.** A radionuclide rebreathing system is a device intended to be used to contain a gaseous or volatile radionuclide or a radionuclide-labeled aerosol and permit it to be respired by the patient during nuclear medicine ventilatory tests (testing process of exchange between the lungs and the atmosphere). This generic type of device may include signal analysis and display equipment, patient and equipment supports, component parts, and accessories.
- (b) **Classification.** Class II.

§ 892.1400 Nuclear sealed calibration source.

- (a) **Identification.** A nuclear sealed calibration source is a device that consists of an encapsulated reference radionuclide intended for calibration of medical nuclear radiation detectors.
- (b) **Classification.** Class I (general controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter, subject to the limitations in § 892.9.

[53 FR 1567, Jan. 20, 1988, as amended at 54 FR 13832, Apr. 5, 1989; 66 FR 38819, July 25, 2001]

§ 892.1410 Nuclear electrocardiograph synchronizer.

- (a) **Identification.** A nuclear electrocardiograph synchronizer is a device intended for use in nuclear radiology to relate the time of image formation to the cardiac cycle during the production of dynamic cardiac images.
- (b) **Classification.** Class I (general controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter subject to § 892.9.

[55 FR 48444, Nov. 20, 1990, as amended at 65 FR 2322, Jan. 14, 2000]

§ 892.1420 Radionuclide test pattern phantom.

- (a) **Identification.** A radionuclide test pattern phantom is a device that consists of an arrangement of radiopaque or radioactive material sealed in a solid pattern intended to serve as a test for a performance characteristic of a nuclear medicine imaging device.
- (b) **Classification.** Class I (general controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter, subject to the limitations in § 892.9.

[53 FR 1567, Jan. 20, 1988, as amended at 54 FR 13832, Apr. 5, 1989; 66 FR 38819, July 25, 2001]

§ 892.1540 Nonfetal ultrasonic monitor.

- (a) **Identification.** A nonfetal ultrasonic monitor is a device that projects a continuous high-frequency sound wave into body tissue other than a fetus to determine frequency changes (doppler shift) in the reflected wave and is intended for use in the investigation of nonfetal blood flow and other nonfetal body tissues in motion. This generic type of device may include signal analysis and display equipment, patient and equipment supports, component parts, and accessories.
- (b) **Classification.** Class II.

§ 892.1550 Ultrasonic pulsed doppler imaging system.

- (a) **Identification.** An ultrasonic pulsed doppler imaging system is a device that combines the features of continuous wave doppler-effect technology with pulsed-echo effect technology and is intended to determine stationary body tissue characteristics, such as depth or location of tissue interfaces or dynamic tissue characteristics such as velocity of blood or tissue motion. This generic type of device may include signal analysis and display equipment, patient and equipment supports, component parts, and accessories.
- (b) **Classification.** Class II.

§ 892.1560 Ultrasonic pulsed echo imaging system.

- (a) **Identification.** An ultrasonic pulsed echo imaging system is a device intended to project a pulsed sound beam into body tissue to determine the depth or location of the tissue interfaces and to measure the duration of an acoustic pulse from the transmitter to the tissue interface and back to the receiver. This generic type of device may include signal analysis and display equipment, patient and equipment supports, component parts, and accessories.
- (b) **Classification.** Class II (special controls). A biopsy needle guide kit intended for use with an ultrasonic pulsed echo imaging system only is exempt from the premarket notification procedures in subpart E of part 807 of this chapter subject to the limitations in § 892.9.

[48 FR 53047, Nov. 23, 1983, as amended at 84 FR 71818, Dec. 30, 2019]

§ 892.1570 Diagnostic ultrasonic transducer.

- (a) **Identification.** A diagnostic ultrasonic transducer is a device made of a piezoelectric material that converts electrical signals into acoustic signals and acoustic signals into electrical signals and intended for use in diagnostic ultrasonic medical devices. Accessories of this generic type of device may include transmission media for acoustically coupling the transducer to the body surface, such as acoustic gel, paste, or a flexible fluid container.
- (b) **Classification.** Class II.

§ 892.1600 Angiographic x-ray system.

- (a) **Identification.** An angiographic x-ray system is a device intended for radiologic visualization of the heart, blood vessels, or lymphatic system during or after injection of a contrast medium. This generic type of device may include signal analysis and display equipment, patient and equipment supports, component parts, and accessories.
- (b) **Classification.** Class II.

§ 892.1610 Diagnostic x-ray beam-limiting device.

- (a) **Identification.** A diagnostic x-ray beam-limiting device is a device such as a collimator, a cone, or an aperture intended to restrict the dimensions of a diagnostic x-ray field by limiting the size of the primary x-ray beam.
- (b) **Classification.** Class II (special controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter subject to the limitations in § 892.9.

[48 FR 53047, Nov. 23, 1983, as amended at 84 FR 71818, Dec. 30, 2019]

§ 892.1620 Cine or spot fluorographic x-ray camera.

- (a) **Identification.** A cine or spot fluorographic x-ray camera is a device intended to photograph diagnostic images produced by x-rays with an image intensifier.
- (b) **Classification.** Class II.

§ 892.1630 Electrostatic x-ray imaging system.

- (a) **Identification.** An electrostatic x-ray imaging system is a device intended for medical purposes that uses an electrostatic field across a semiconductive plate, a gas-filled chamber, or other similar device to convert a pattern of x-radiation into an electrostatic image and, subsequently, into a visible image. This generic type of device may include signal analysis and display equipment, patient and equipment supports, component parts, and accessories.
- (b) **Classification.** Class II.

§ 892.1640 Radiographic film marking system.

- (a) **Identification.** A radiographic film marking system is a device intended for medical purposes to add identification and other information onto radiographic film by means of exposure to visible light.
- (b) **Classification.** Class I (general controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter, subject to the limitations in § 892.9.

[55 FR 48444, Nov. 20, 1990, as amended at 59 FR 63015, Dec. 7, 1994; 66 FR 38819, July 25, 2001]

§ 892.1650 Image-intensified fluoroscopic x-ray system.

- (a) **Identification.** An image-intensified fluoroscopic x-ray system is a device intended to visualize anatomical structures by converting a pattern of x-radiation into a visible image through electronic amplification. This generic type of device may include signal analysis and display equipment, patient and equipment supports, component parts, and accessories.
- (b) **Classification.** Class II (special controls). An anthropogram tray or radiology dental tray intended for use with an image-intensified fluoroscopic x-ray system only is exempt from the premarket notification procedures in subpart E of part 807 of this chapter subject to the limitations in § 892.9. In addition, when intended as an accessory to the device described in paragraph (a) of this section, the fluoroscopic compression device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter subject to the limitations in § 892.9.

[53 FR 1567, Jan. 20, 1988, as amended at 66 FR 57369, Nov. 15, 2001; 84 FR 71818, Dec. 30, 2019]

§ 892.1660 Non-image-intensified fluoroscopic x-ray system.

- (a) **Identification.** A non-image-intensified fluoroscopic x-ray system is a device intended to be used to visualize anatomical structures by using a fluorescent screen to convert a pattern of x-radiation into a visible image. This generic type of device may include signal analysis and display equipment, patient and equipment supports, component parts, and accessories.
- (b) **Classification.** Class II.

§ 892.1670 Spot-film device.

- (a) **Identification.** A spot-film device is an electromechanical component of a fluoroscopic x-ray system that is intended to be used for medical purposes to position a radiographic film cassette to obtain radiographs during fluoroscopy.
- (b) **Classification.** Class II (special controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter subject to the limitations in § 892.9.

[48 FR 53047, Nov. 23, 1983, as amended at 84 FR 71818, Dec. 30, 2019]

§ 892.1680 Stationary x-ray system.

- (a) **Identification.** A stationary x-ray system is a permanently installed diagnostic system intended to generate and control x-rays for examination of various anatomical regions. This generic type of device may include signal analysis and display equipment, patient and equipment supports, component parts, and accessories.
- (b) **Classification.** Class II (special controls). A radiographic contrast tray or radiology diagnostic kit intended for use with a stationary x-ray system only is exempt from the premarket notification procedures in subpart E of part 807 of this chapter subject to the limitations in § 892.9.

[48 FR 53047, Nov. 23, 1983, as amended at 84 FR 71818, Dec. 30, 2019]

§ 892.1700 Diagnostic x-ray high voltage generator.

- (a) **Identification.** A diagnostic x-ray high voltage generator is a device that is intended to supply and control the electrical energy applied to a diagnostic x-ray tube for medical purposes. This generic type of device may include a converter that changes alternating current to direct current, filament transformers for the x-ray tube, high voltage switches, electrical protective devices, or other appropriate elements.
- (b) **Classification.** Class I (general controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter, subject to the limitations in § 892.9.

[53 FR 1567, Jan. 20, 1988, as amended at 61 FR 1125, Jan. 16, 1996; 66 FR 38819, July 25, 2001]

§ 892.1710 Mammographic x-ray system.

- (a) **Identification.** A mammographic x-ray system is a device intended to be used to produce radiographs of the breast. This generic type of device may include signal analysis and display equipment, patient and equipment supports, component parts, and accessories.
- (b) **Classification.** Class II.

§ 892.1715 Full-field digital mammography system.

- (a) **Identification.** A full-field digital mammography system is a device intended to produce planar digital x-ray images of the entire breast. This generic type of device may include digital mammography acquisition software, full-field digital image receptor, acquisition workstation, automatic exposure control, image processing and reconstruction programs, patient and equipment supports, component parts, and accessories.
- (b) **Classification.** Class II (special controls). The special control for the device is FDA's guidance document entitled "Class II Special Controls Guidance Document: Full-Field Digital Mammography System." See § 892.1(e) for the availability of this guidance document.

[75 FR 68203, Nov. 5, 2010]

§ 892.1720 Mobile x-ray system.

- (a) **Identification.** A mobile x-ray system is a transportable device system intended to be used to generate and control x-ray for diagnostic procedures. This generic type of device may include signal analysis and display equipment, patient and equipment supports, component parts, and accessories.
- (b) **Classification.** Class II.

§ 892.1730 Photofluorographic x-ray system.

- (a) **Identification.** A photofluorographic x-ray system is a device that includes a fluoroscopic x-ray unit and a camera intended to be used to produce, then photograph, a fluoroscopic image of the body. This generic type of device may include signal analysis and display equipment, patient and equipment supports, component parts, and accessories.
- (b) **Classification.** Class II (special controls). A discography kit intended for use with a photofluorographic x-ray system only is exempt from the premarket notification procedures in subpart E of part 807 of this chapter subject to the limitations in § 892.9.

[48 FR 53047, Nov. 23, 1983, as amended at 84 FR 71819, Dec. 30, 2019]

§ 892.1740 Tomographic x-ray system.

- (a) **Identification.** A tomographic x-ray system is an x-ray device intended to be used to produce radiologic images of a specific cross-sectional plane of the body by blurring or eliminating detail from other planes. This generic type of device may include signal analysis and display equipment, patient and equipment supports, component parts, and accessories.
- (b) **Classification.** Class II.

§ 892.1750 Computed tomography x-ray system.

- (a) **Identification.** A computed tomography x-ray system is a diagnostic x-ray system intended to produce cross-sectional images of the body by computer reconstruction of x-ray transmission data from the same axial plane taken at different angles. This generic type of device may include signal analysis and display equipment, patient and equipment supports, component parts, and accessories.
- (b) **Classification.** Class II.

§ 892.1760 Diagnostic x-ray tube housing assembly.

- (a) **Identification.** A diagnostic x-ray tube housing assembly is an x-ray generating tube encased in a radiation-shielded housing that is intended for diagnostic purposes. This generic type of device may include high voltage and filament transformers or other appropriate components.
- (b) **Classification.** Class I (general controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter, subject to the limitations in § 892.9.

[53 FR 1567, Jan. 20, 1988, as amended at 61 FR 1125, Jan. 16, 1996; 66 FR 38819, July 25, 2001]

§ 892.1770 Diagnostic x-ray tube mount.

- (a) **Identification.** A diagnostic x-ray tube mount is a device intended to support and to position the diagnostic x-ray tube housing assembly for a medical radiographic procedure.
- (b) **Classification.** Class I (general controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter, subject to the limitations in § 892.9.

[53 FR 1567, Jan. 20, 1988, as amended at 61 FR 1125, Jan. 16, 1996; 66 FR 38819, July 25, 2001]

§ 892.1820 Pneumoencephalographic chair.

- (a) **Identification.** A pneumoencephalographic chair is a chair intended to support and position a patient during pneumoencephalography (x-ray imaging of the brain).
- (b) **Classification.** Class II (special controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter subject to the limitations in § 892.9.

[48 FR 53047, Nov. 23, 1983, as amended at 84 FR 71819, Dec. 30, 2019]

§ 892.1830 Radiologic patient cradle.

- (a) **Identification.** A radiologic patient cradle is a support device intended to be used for rotational positioning about the longitudinal axis of a patient during radiologic procedures.
- (b) **Classification.** Class I (general controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter, subject to the limitations in § 892.9.

[53 FR 1567, Jan. 20, 1988, as amended at 61 FR 1125, Jan. 16, 1996; 66 FR 38819, July 25, 2001]

§ 892.1840 Radiographic film.

- (a) **Identification.** Radiographic film is a device that consists of a thin sheet of radiotransparent material coated on one or both sides with a photographic emulsion intended to record images during diagnostic radiologic procedures.
- (b) **Classification.** Class I (general controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter, subject to the limitations in § 892.9.

[53 FR 1567, Jan. 20, 1988, as amended at 66 FR 38819, July 25, 2001]

§ 892.1850 Radiographic film cassette.

- (a) **Identification.** A radiographic film cassette is a device intended for use during diagnostic x-ray procedures to hold a radiographic film in close contact with an x-ray intensifying screen and to provide a light-proof enclosure for direct exposure of radiographic film.
- (b) **Classification.** Class II (special controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter subject to the limitations in § 892.9.

[48 FR 53047, Nov. 23, 1983, as amended at 84 FR 71819, Dec. 30, 2019]

§ 892.1860 Radiographic film/cassette changer.

- (a) **Identification.** A radiographic film/cassette changer is a device intended to be used during a radiologic procedure to move a radiographic film or cassette between x-ray exposures and to position it during the exposure.
- (b) **Classification.** Class II (special controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter subject to the limitations in § 892.9.

[48 FR 53047, Nov. 23, 1983, as amended at 84 FR 71819, Dec. 30, 2019]

§ 892.1870 Radiographic film/cassette changer programmer.

- (a) **Identification.** A radiographic film/cassette changer programmer is a device intended to be used to control the operations of a film or cassette changer during serial medical radiography.
- (b) **Classification.** Class II (special controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter subject to the limitations in § 892.9.

[48 FR 53047, Nov. 23, 1983, as amended at 84 FR 71819, Dec. 30, 2019]

§ 892.1880 Wall-mounted radiographic cassette holder.

- (a) **Identification.** A wall-mounted radiographic cassette holder is a device that is a support intended to hold and position radiographic cassettes for a radiographic exposure for medical use.
- (b) **Classification.** Class I (general controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter, subject to the limitations in § 892.9.

[53 FR 1567, Jan. 20, 1988, as amended at 61 FR 1125, Jan. 16, 1996; 66 FR 38819, July 25, 2001]

§ 892.1890 Radiographic film illuminator.

- (a) **Identification.** A radiographic film illuminator is a device containing a visible light source covered with a translucent front that is intended to be used to view medical radiographs.
- (b) **Classification.** Class I (general controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter subject to § 892.9.

[55 FR 48444, Nov. 20, 1990, as amended at 65 FR 2323, Jan. 14, 2000]

§ 892.1900 Automatic radiographic film processor.

- (a) **Identification.** An automatic radiographic film processor is a device intended to be used to develop, fix, wash, and dry automatically and continuously film exposed for medical purposes.
- (b) **Classification.** Class II (special controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter subject to the limitations in § 892.9.

[55 FR 48444, Nov. 20, 1990, as amended at 84 FR 71819, Dec. 30, 2019]

§ 892.1910 Radiographic grid.

- (a) **Identification.** A radiographic grid is a device that consists of alternating radiolucent and radiopaque strips intended to be placed between the patient and the image receptor to reduce the amount of scattered radiation reaching the image receptor.
- (b) **Classification.** Class I (general controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter subject to § 892.9.

[53 FR 1567, Jan. 20, 1988, as amended at 65 FR 2323, Jan. 14, 2000]

§ 892.1920 Radiographic head holder.

- (a) **Identification.** A radiographic head holder is a device intended to position the patient's head during a radiographic procedure.
- (b) **Classification.** Class I (general controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter, subject to the limitations in § 892.9. The device is also exempt from the current good manufacturing practice requirements of the quality system regulation in part 820 of this chapter, with the exception of § 820.180, with respect to general requirements concerning records, and § 820.198, with respect to complaint files.

[53 FR 1567, Jan. 20, 1988, as amended at 66 FR 38819, July 25, 2001]

§ 892.1940 Radiologic quality assurance instrument.

- (a) **Identification.** A radiologic quality assurance instrument is a device intended for medical purposes to measure a physical characteristic associated with another radiologic device.
- (b) **Classification.** Class I (general controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter, subject to the limitations in § 892.9. The device is also exempt from the current good manufacturing practice requirements of the quality system regulation in part 820 of this chapter, with the exception of § 820.180, with respect to general requirements concerning records, and § 820.198, with respect to complaint files.

[53 FR 1567, Jan. 20, 1988, as amended at 66 FR 38819, July 25, 2001]

§ 892.1950 Radiographic anthropomorphic phantom.

- (a) **Identification.** A radiographic anthropomorphic phantom is a device intended for medical purposes to simulate a human body for positioning radiographic equipment.
- (b) **Classification.** Class I (general controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter, subject to the limitations in § 892.9. The device is also exempt from the current good manufacturing practice requirements of the quality system regulation in part 820 of this chapter, with the exception of § 820.180, with respect to general requirements concerning records, and § 820.198, with respect to complaint files.

[53 FR 1567, Jan. 20, 1988, as amended at 66 FR 38819, July 25, 2001]

§ 892.1960 Radiographic intensifying screen.

- (a) **Identification.** A radiographic intensifying screen is a device that is a thin radiolucent sheet coated with a luminescent material that transforms incident x-ray photons into visible light and intended for medical purposes to expose radiographic film.
- (b) **Classification.** Class I (general controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter subject to § 892.9.

[53 FR 1567, Jan. 20, 1988, as amended at 65 FR 2323, Jan. 14, 2000]

§ 892.1970 Radiographic ECG/respirator synchronizer.

- (a) **Identification.** A radiographic ECG/respirator synchronizer is a device intended to be used to coordinate an x-ray film exposure with the signal from an electrocardiograph (ECG) or respirator at a predetermined phase of the cardiac or respiratory cycle.
- (b) **Classification.** Class I (general controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter subject to § 892.9.

[55 FR 48444, Nov. 20, 1990, as amended at 65 FR 2323, Jan. 14, 2000]

§ 892.1980 Radiologic table.

- (a) **Identification.** A radiologic table is a device intended for medical purposes to support a patient during radiologic procedures. The table may be fixed or tilting and may be electrically powered.
- (b) **Classification.** Class II (special controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter subject to § 892.9.

[53 FR 1567, Jan. 20, 1988, as amended at 63 FR 59231, Nov. 3, 1998]

§ 892.1990 Transilluminator for breast evaluation.

- (a) **Identification.** A transilluminator, also known as a diaphanoscope or lightscanner, is an electrically powered device that uses low intensity emissions of visible light and near-infrared radiation (approximately 700-1050 nanometers (nm)), transmitted through the breast, to visualize translucent tissue for the diagnosis of cancer, other conditions, diseases, or abnormalities.
- (b) **Classification.** Class III (premarket approval).
- (c) **Date premarket approval (PMA) or notice of completion of product development protocol (PDP) is required.** A PMA or notice of completion of a PDP is required to be filed with FDA by April 17, 2014, for any transilluminator for breast evaluation that was in commercial distribution before May 28, 1976, or that has, by April 17, 2014, been found to be substantially equivalent to any transilluminator for breast evaluation that was in commercial distribution before May 28, 1976. Any other transilluminator for breast evaluation shall have an approved PMA or declared completed PDP in effect before being placed in commercial distribution.

[60 FR 36639, July 18, 1995, as amended at 79 FR 3094, Jan. 17, 2014]

§ 892.2010 Medical image storage device.

- (a) **Identification.** A medical image storage device is a hardware device that provides electronic storage and retrieval functions for medical images. Examples include electronic hardware devices employing magnetic and optical discs, magnetic tapes, and digital memory.
- (b) **Classification.** Class I (general controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter subject to § 892.9.

[63 FR 23387, Apr. 29, 1998; 63 FR 44998, Aug. 24, 1998, as amended at 65 FR 2323, Jan. 14, 2000; 86 FR 20284, Apr. 19, 2021]

§ 892.2020 Medical image communications device.

- (a) **Identification.** A medical image communications device provides electronic transfer of medical image data between medical devices. It may include a physical communications medium, modems, and interfaces. It may provide simple image review software functionality for medical image processing and manipulation, such as grayscale window and level, zoom and pan, user delineated geometric measurements, compression, or user added image annotations. The device does not perform advanced image processing or complex quantitative functions. This does not include electronic transfer of medical image software functions.
- (b) **Classification.** Class I (general controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter subject to § 892.9.

[63 FR 23387, Apr. 29, 1998; 63 FR 44998, Aug. 24, 1998, as amended at 65 FR 2323, Jan. 14, 2000; 86 FR 20283, Apr. 19, 2021]

§ 892.2030 Medical image digitizer.

- (a) **Identification.** A medical image digitizer is a device intended to convert an analog medical image into a digital format. Examples include systems employing video frame grabbers, and scanners which use lasers or charge-coupled devices.
- (b) **Classification.** Class II (special controls; voluntary standards—Digital Imaging and Communications in Medicine (DICOM) Std., Joint Photographic Experts Group (JPEG) Std.). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter subject to the limitations in § 892.9.

[63 FR 23387, Apr. 29, 1998, as amended at 84 FR 71819, Dec. 30, 2019]

§ 892.2040 Medical image hardcopy device.

- (a) **Identification.** A medical image hardcopy device is a device that produces a visible printed record of a medical image and associated identification information. Examples include multifunction cameras and laser printers.
- (b) **Classification.** Class II (special controls; voluntary standards—Digital Imaging and Communications in Medicine (DICOM) Std., Joint Photographic Experts Group (JPEG) Std., Society of Motion Picture and Television Engineers (SMPTE) Test Pattern). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter subject to the limitations in § 892.9.

[63 FR 23387, Apr. 29, 1998, as amended at 84 FR 71819, Dec. 30, 2019]

§ 892.2050 Medical image management and processing system.

- (a) **Identification.** A medical image management and processing system is a device that provides one or more capabilities relating to the review and digital processing of medical images for the purposes of interpretation by a trained practitioner of disease detection, diagnosis, or patient management. The software components may provide advanced or complex image processing functions for image manipulation, enhancement, or quantification that are intended for use in the interpretation and analysis of medical images. Advanced image manipulation functions may include image segmentation, multimodality image registration, or 3D visualization. Complex quantitative functions may include semi-automated measurements or time-series measurements.
- (b) **Classification.** Class II (special controls; voluntary standards—Digital Imaging and Communications in Medicine (DICOM) Std., Joint Photographic Experts Group (JPEG) Std., Society of Motion Picture and Television Engineers (SMPTE) Test Pattern).

[63 FR 23387, Apr. 29, 1998, as amended at 86 FR 20284, Apr. 19, 2021]

§ 892.2060 Radiological computer-assisted diagnostic software for lesions suspicious of cancer.

- (a) **Identification.** A radiological computer-assisted diagnostic software for lesions suspicious of cancer is an image processing prescription device intended to aid in the characterization of lesions as suspicious for cancer identified on acquired medical images such as magnetic resonance, mammography, radiography, or computed tomography. The device characterizes lesions based on features or information extracted from the images and provides information about the lesion(s) to the user. Diagnostic and patient management decisions are made by the clinical user.
- (b) **Classification.** Class II (special controls). The special controls for this device are:
 - (1) Design verification and validation must include:
 - (i) A detailed description of the image analysis algorithms including, but not limited to, a detailed description of the algorithm inputs and outputs, each major component or block, and algorithm limitations.
 - (ii) A detailed description of pre-specified performance testing protocols and dataset(s) used to assess whether the device will improve reader performance as intended.
 - (iii) Results from performance testing protocols that demonstrate that the device improves reader performance in the intended use population when used in accordance with the instructions for use. The performance assessment must be based on appropriate diagnostic accuracy measures (e.g., receiver operator characteristic plot, sensitivity, specificity, predictive value, and diagnostic likelihood ratio). The test dataset must contain sufficient numbers of cases from important cohorts (e.g., subsets defined by clinically relevant confounders, effect modifiers, concomitant diseases, and subsets defined by image acquisition characteristics) such that the performance estimates and confidence intervals of the device for these individual subsets can be characterized for the intended use population and imaging equipment.
 - (iv) Standalone performance testing protocols and results of the device.

- (v) Appropriate software documentation (e.g., device hazard analysis; software requirements specification document; software design specification document; traceability analysis; and description of verification and validation activities including system level test protocol, pass/fail criteria, results, and cybersecurity).
- (2) Labeling must include:
 - (i) A detailed description of the patient population for which the device is indicated for use.
 - (ii) A detailed description of the intended reading protocol.
 - (iii) A detailed description of the intended user and recommended user training.
 - (iv) A detailed description of the device inputs and outputs.
 - (v) A detailed description of compatible imaging hardware and imaging protocols.
 - (vi) Warnings, precautions, and limitations, including situations in which the device may fail or may not operate at its expected performance level (e.g., poor image quality or for certain subpopulations), as applicable.
 - (vii) Detailed instructions for use.
 - (viii) A detailed summary of the performance testing, including: Test methods, dataset characteristics, results, and a summary of sub-analyses on case distributions stratified by relevant confounders (e.g., lesion and organ characteristics, disease stages, and imaging equipment).

[85 FR 3542, Jan. 22, 2020]

§ 892.2070 Medical image analyzer.

- (a) **Identification.** Medical image analyzers, including computer-assisted/aided detection (CADe) devices for mammography breast cancer, ultrasound breast lesions, radiograph lung nodules, and radiograph dental caries detection, is a prescription device that is intended to identify, mark, highlight, or in any other manner direct the clinicians' attention to portions of a radiology image that may reveal abnormalities during interpretation of patient radiology images by the clinicians. This device incorporates pattern recognition and data analysis capabilities and operates on previously acquired medical images. This device is not intended to replace the review by a qualified radiologist, and is not intended to be used for triage, or to recommend diagnosis.
- (b) **Classification.** Class II (special controls). The special controls for this device are:
 - (1) Design verification and validation must include:
 - (i) A detailed description of the image analysis algorithms including a description of the algorithm inputs and outputs, each major component or block, and algorithm limitations.
 - (ii) A detailed description of pre-specified performance testing methods and dataset(s) used to assess whether the device will improve reader performance as intended and to characterize the standalone device performance. Performance testing includes one or more standalone tests, side-by-side comparisons, or a reader study, as applicable.

- (iii) Results from performance testing that demonstrate that the device improves reader performance in the intended use population when used in accordance with the instructions for use. The performance assessment must be based on appropriate diagnostic accuracy measures (e.g., receiver operator characteristic plot, sensitivity, specificity, predictive value, and diagnostic likelihood ratio). The test dataset must contain a sufficient number of cases from important cohorts (e.g., subsets defined by clinically relevant confounders, effect modifiers, concomitant diseases, and subsets defined by image acquisition characteristics) such that the performance estimates and confidence intervals of the device for these individual subsets can be characterized for the intended use population and imaging equipment.
 - (iv) Appropriate software documentation (e.g., device hazard analysis; software requirements specification document; software design specification document; traceability analysis; description of verification and validation activities including system level test protocol, pass/fail criteria, and results; and cybersecurity).
- (2) Labeling must include the following:
- (i) A detailed description of the patient population for which the device is indicated for use.
 - (ii) A detailed description of the intended reading protocol.
 - (iii) A detailed description of the intended user and user training that addresses appropriate reading protocols for the device.
 - (iv) A detailed description of the device inputs and outputs.
 - (v) A detailed description of compatible imaging hardware and imaging protocols.
 - (vi) Discussion of warnings, precautions, and limitations must include situations in which the device may fail or may not operate at its expected performance level (e.g., poor image quality or for certain subpopulations), as applicable.
 - (vii) Device operating instructions.
 - (viii) A detailed summary of the performance testing, including: test methods, dataset characteristics, results, and a summary of sub-analyses on case distributions stratified by relevant confounders, such as lesion and organ characteristics, disease stages, and imaging equipment.

[85 FR 3548, Jan. 22, 2020]

§ 892.2080 Radiological computer aided triage and notification software.

- (a) **Identification.** Radiological computer aided triage and notification software is an image processing prescription device intended to aid in prioritization and triage of radiological medical images. The device notifies a designated list of clinicians of the availability of time sensitive radiological medical images for review based on computer aided image analysis of those images performed by the device. The device does not mark, highlight, or direct users' attention to a specific location in the original image. The device does not remove cases from a reading queue. The device operates in parallel with the standard of care, which remains the default option for all cases.
- (b) **Classification.** Class II (special controls). The special controls for this device are:
 - (1) Design verification and validation must include:

- (i) A detailed description of the notification and triage algorithms and all underlying image analysis algorithms including, but not limited to, a detailed description of the algorithm inputs and outputs, each major component or block, how the algorithm affects or relates to clinical practice or patient care, and any algorithm limitations.
 - (ii) A detailed description of pre-specified performance testing protocols and dataset(s) used to assess whether the device will provide effective triage (e.g., improved time to review of prioritized images for pre-specified clinicians).
 - (iii) Results from performance testing that demonstrate that the device will provide effective triage. The performance assessment must be based on an appropriate measure to estimate the clinical effectiveness. The test dataset must contain sufficient numbers of cases from important cohorts (e.g., subsets defined by clinically relevant confounders, effect modifiers, associated diseases, and subsets defined by image acquisition characteristics) such that the performance estimates and confidence intervals for these individual subsets can be characterized with the device for the intended use population and imaging equipment.
 - (iv) Stand-alone performance testing protocols and results of the device.
 - (v) Appropriate software documentation (e.g., device hazard analysis; software requirements specification document; software design specification document; traceability analysis; description of verification and validation activities including system level test protocol, pass/fail criteria, and results).
- (2) Labeling must include the following:
- (i) A detailed description of the patient population for which the device is indicated for use;
 - (ii) A detailed description of the intended user and user training that addresses appropriate use protocols for the device;
 - (iii) Discussion of warnings, precautions, and limitations must include situations in which the device may fail or may not operate at its expected performance level (e.g., poor image quality for certain subpopulations), as applicable;
 - (iv) A detailed description of compatible imaging hardware, imaging protocols, and requirements for input images;
 - (v) Device operating instructions; and
 - (vi) A detailed summary of the performance testing, including: test methods, dataset characteristics, triage effectiveness (e.g., improved time to review of prioritized images for pre-specified clinicians), diagnostic accuracy of algorithms informing triage decision, and results with associated statistical uncertainty (e.g., confidence intervals), including a summary of subanalyses on case distributions stratified by relevant confounders, such as lesion and organ characteristics, disease stages, and imaging equipment.

[85 FR 3544, Jan. 22, 2020]

§ 892.2090 Radiological computer-assisted detection and diagnosis software.

- (a) **Identification.** A radiological computer-assisted detection and diagnostic software is an image processing device intended to aid in the detection, localization, and characterization of fracture, lesions, or other disease-specific findings on acquired medical images (e.g., radiography, magnetic resonance, computed tomography). The device detects, identifies, and characterizes findings based on features or information extracted from images, and provides information about the presence, location, and characteristics of the findings to the user. The analysis is intended to inform the primary diagnostic and patient management decisions that are made by the clinical user. The device is not intended as a replacement for a complete clinician's review or their clinical judgment that takes into account other relevant information from the image or patient history.
- (b) **Classification.** Class II (special controls). The special controls for this device are:
 - (1) Design verification and validation must include:
 - (i) A detailed description of the image analysis algorithm, including a description of the algorithm inputs and outputs, each major component or block, how the algorithm and output affects or relates to clinical practice or patient care, and any algorithm limitations.
 - (ii) A detailed description of pre-specified performance testing protocols and dataset(s) used to assess whether the device will provide improved assisted-read detection and diagnostic performance as intended in the indicated user population(s), and to characterize the standalone device performance for labeling. Performance testing includes standalone test(s), side-by-side comparison(s), and/or a reader study, as applicable.
 - (iii) Results from standalone performance testing used to characterize the independent performance of the device separate from aided user performance. The performance assessment must be based on appropriate diagnostic accuracy measures (e.g., receiver operator characteristic plot, sensitivity, specificity, positive and negative predictive values, and diagnostic likelihood ratio). Devices with localization output must include localization accuracy testing as a component of standalone testing. The test dataset must be representative of the typical patient population with enrichment made only to ensure that the test dataset contains a sufficient number of cases from important cohorts (e.g., subsets defined by clinically relevant confounders, effect modifiers, concomitant disease, and subsets defined by image acquisition characteristics) such that the performance estimates and confidence intervals of the device for these individual subsets can be characterized for the intended use population and imaging equipment.
 - (iv) Results from performance testing that demonstrate that the device provides improved assisted-read detection and/or diagnostic performance as intended in the indicated user population(s) when used in accordance with the instructions for use. The reader population must be comprised of the intended user population in terms of clinical training, certification, and years of experience. The performance assessment must be based on appropriate diagnostic accuracy measures (e.g., receiver operator characteristic plot, sensitivity, specificity, positive and negative predictive values, and diagnostic likelihood ratio). Test datasets must meet the requirements described in paragraph (b)(1)(iii) of this section.
 - (v) Appropriate software documentation, including device hazard analysis, software requirements specification document, software design specification document, traceability analysis, system level test protocol, pass/fail criteria, testing results, and cybersecurity measures.

(2) Labeling must include the following:

- (i) A detailed description of the patient population for which the device is indicated for use.
- (ii) A detailed description of the device instructions for use, including the intended reading protocol and how the user should interpret the device output.
- (iii) A detailed description of the intended user, and any user training materials or programs that address appropriate reading protocols for the device, to ensure that the end user is fully aware of how to interpret and apply the device output.
- (iv) A detailed description of the device inputs and outputs.
- (v) A detailed description of compatible imaging hardware and imaging protocols.
- (vi) Warnings, precautions, and limitations must include situations in which the device may fail or may not operate at its expected performance level (e.g., poor image quality or for certain subpopulations), as applicable.
- (vii) A detailed summary of the performance testing, including test methods, dataset characteristics, results, and a summary of sub-analyses on case distributions stratified by relevant confounders, such as anatomical characteristics, patient demographics and medical history, user experience, and imaging equipment.

[90 FR 24970, June 13, 2025]

892.2100 Radiological acquisition and/or optimization guidance system.

892.2100 Radiological acquisition and/or optimization guidance system.

- (a) **Identification.** A radiological acquisition and/or optimization guidance system is a device that is intended to aid in the acquisition and/or optimization of images and/or diagnostic signals. The device interfaces with the acquisition system, analyzes its output, and provides guidance and/or feedback to the operator for improving image and/or signal quality.
- (b) **Classification.** Class II (special controls). The special controls for this device are:
 - (1) Design verification and validation must include:
 - (i) A detailed, technical device description, including a detailed description of the impact of any software and hardware on the device's functions, the associated capabilities and limitations of each part, and the associated inputs and outputs.
 - (ii) A detailed, technical report on the non-clinical performance testing of the subject device in the intended use environments, using relevant consensus standards when applicable.
 - (iii) A detailed report on the clinical performance testing, obtained from either clinical testing, accepted virtual/physical systems designed to capture clinical variability, comparison to a closely-related device with established clinical performance, or other sources that are justified appropriately. The choice of the method must be justified given the risk of the device and the general acceptance of the test methods. The report must include the following:
 - (A) A thorough description of the testing protocol(s).

- (B) A thorough, quantitative evaluation of the diagnostic utility and quality of images/data acquired, or optimized, using the device.
 - (C) A thorough, quantitative evaluation of the performance in a representative user population and patient population, under anticipated conditions and environments of use.
 - (D) A thorough discussion on the generalizability of the clinical performance testing results.
 - (E) A thorough discussion on use-related risk analysis/human factors data.
 - (iv) A detailed protocol that describes, in the event of a future change, the level of change in the device technical specifications or indications for use at which the change or changes could significantly affect the safety or effectiveness of the device and the risks posed by these changes. The assessment metrics, acceptance criteria, and analytical methods used for the performance testing of changes that are within the scope of the protocol must be included.
 - (v) Documentation of an appropriate training program, including instructions on how to acquire and process quality images and video clips, and a report on usability testing demonstrating the effectiveness of that training program on user performance, including acquiring and processing quality images.
- (2) The labeling required under § 801.109(c) of this chapter must include:
- (i) A detailed description of the device, including information on all required and/or compatible parts.
 - (ii) A detailed description of the patient population for which the device is indicated for use.
 - (iii) A detailed description of the intended user population, and the recommended user training.
 - (iv) Detailed instructions for use, including the information provided in the training program used to meet the requirements of paragraph (b)(1)(iv) of this section.
 - (v) A warning that the images and data acquired using the device are to be interpreted only by qualified medical professionals.
 - (vi) A detailed summary of the reports required under paragraphs (b)(1)(ii) and (iii) of this section.
 - (vii) A statement on upholding the As Low As Reasonably Achievable (ALARA) principle with a discussion on the associated device controls/options.

[90 FR 23285, June 2, 2025]

Subparts C-E [Reserved]

Subpart F—Therapeutic Devices

§ 892.5050 Medical charged-particle radiation therapy system.

- (a) **Identification.** A medical charged-particle radiation therapy system is a device that produces by acceleration high energy charged particles (e.g., electrons and protons) intended for use in radiation therapy. This generic type of device may include signal analysis and display equipment, patient and equipment supports, treatment planning computer programs, component parts, and accessories.

- (b) **Classification.** Class II. When intended for use as a quality control system, the film dosimetry system (film scanning system) included as an accessory to the device described in paragraph (a) of this section, is exempt from the premarket notification procedures in subpart E of part 807 of this chapter subject to the limitations in § 892.9.

[53 FR 1567, Jan. 20, 1988, as amended at 64 FR 1125, Jan. 8, 1999]

§ 892.5300 Medical neutron radiation therapy system.

- (a) **Identification.** A medical neutron radiation therapy system is a device intended to generate high-energy neutrons for radiation therapy. This generic type of device may include signal analysis and display equipment, patient and equipment support, treatment planning computer programs, component parts, and accessories.
- (b) **Classification.** Class II.

§ 892.5650 Manual radionuclide applicator system.

- (a) **Identification.** A manual radionuclide applicator system is a manually operated device intended to apply a radionuclide source into the body or to the surface of the body for radiation therapy. This generic type of device may include patient and equipment supports, component parts, treatment planning computer programs, and accessories.
- (b) **Classification.** Class I (general controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter subject to § 892.9.

[53 FR 1567, Jan. 20, 1988, as amended at 65 FR 2323, Jan. 14, 2000]

§ 892.5700 Remote controlled radionuclide applicator system.

- (a) **Identification.** A remote controlled radionuclide applicator system is an electromechanical or pneumatic device intended to enable an operator to apply, by remote control, a radionuclide source into the body or to the surface of the body for radiation therapy. This generic type of device may include patient and equipment supports, component parts, treatment planning computer programs, and accessories.
- (b) **Classification.** Class II.

§ 892.5710 Radiation therapy beam-shaping block.

- (a) **Identification.** A radiation therapy beam-shaping block is a device made of a highly attenuating material (such as lead) intended for medical purposes to modify the shape of a beam from a radiation therapy source.
- (b) **Classification.** Class II.

§ 892.5720 Rectal balloon for prostate immobilization.

- (a) **Identification.** A rectal balloon for prostate immobilization is a single use, inflatable, non-powered positioning device placed in the rectum to immobilize the prostate in patients undergoing radiation therapy. The device is intended to be used during all the phases of radiation therapy, including treatment planning, image verification, and radiotherapy delivery.
- (b) **Classification.** Class II (special controls). The special controls for this device are:

- (1) The premarket notification submission must include methodology and results of the following non-clinical and clinical performance testing:
 - (i) Biocompatibility testing of the final finished device;
 - (ii) If provided sterile, sterilization validation;
 - (iii) If not provided sterile, bioburden testing of the final finished device;
 - (iv) Shelf life and expiration date validation; and
 - (v) Performance testing including but not limited to:
 - (A) Venting mechanism (if device has a vent mechanism);
 - (B) Safety mechanism(s) to prevent advancement beyond its intended safe placement; and
 - (C) Structural integrity testing (e.g., tensile strength, balloon leakage and burst strength).
- (2) Labeling that includes:
 - (i) Appropriate warnings and contraindications, including, but not limited to the following statements:
 - (A) "Do not transport the patient with the rectal balloon inserted. The balloon should be removed prior to transport.";
 - (B) "Failure to perform the standard imaging position verification protocol may cause the device to not perform as intended.";
 - (C) "Reduce the rectal balloon fill volume if the patient experiences discomfort due to the rectal balloon inflation."; and
 - (D) "Do not apply excessive pressure/force on the shaft or tubing of the rectal balloon."
 - (ii) Adequate instructions for use on the proper insertion procedure, positioning, and inflation of the rectal balloon;
 - (iii) Whether the device is sterile or non-sterile; and
 - (iv) An expiration date.

[82 FR 61171, Dec. 27, 2017]

§ 892.5725 Absorbable perirectal spacer.

- (a) **Identification.** An absorbable perirectal spacer is composed of biodegradable material that temporarily positions the anterior rectal wall away from the prostate during radiotherapy for prostate cancer with the intent to reduce the radiation dose delivered to the anterior rectum. The absorbable spacer maintains space for the entire course of prostate radiotherapy treatment and is completely absorbed by the patient's body over time.
- (b) **Classification.** Class II (special controls). The special controls for this device are:
 - (1) The premarket notification submission must include methodology and results of the following non-clinical and clinical performance testing. For all clinical investigations used to support premarket notification submissions for this type of device, line listings of the study data must be provided.

- (i) Performance bench testing must demonstrate appropriate perirectal space creation and maintenance for the duration of prostate radiotherapy.
 - (ii) Performance bench testing must demonstrate that therapeutic radiation levels do not alter the performance of the device.
 - (iii) Performance in vivo testing must demonstrate appropriate deployment of spacer as indicated in the accompanying labeling, and demonstrate appropriate expansion and absorption characteristics in a clinically relevant environment.
 - (iv) Clinical study must demonstrate appropriate spacer stability and lack of migration for the entire course of radiotherapy, complete absorption, and lack of long term toxicity.
 - (v) Sterility testing must demonstrate the sterility of the device and the effects of the sterilization process on the physical characteristics of the spacer.
 - (vi) Shelf-life testing must demonstrate the stability of the physical characteristics of the spacer throughout the shelf-life as indicated in the accompanying labeling.
 - (vii) The device must be demonstrated to be biocompatible.
- (2) The risk management activities performed as part of the manufacturer's § 820.30 design controls must document an appropriate end user initial training program which will be offered as part of efforts to mitigate the risk of failure to correctly operate the device, including, but not limited to, documentation of an appropriate end user initial training program on the proper spacer deployment technique.
- (3) The device labeling must include the following:
- (i) A detailed summary of reported or observed complications related to the use of the device;
 - (ii) Appropriate warnings;
 - (iii) Detailed instructions for system preparations and detailed implant procedure instructions; and
 - (iv) An expiration date that is supported by performance data as specified in paragraph (b)(1)(vi) of this section.

[83 FR 601, Jan. 5, 2018]

§ 892.5730 Radionuclide brachytherapy source.

- (a) **Identification.** A radionuclide brachytherapy source is a device that consists of a radionuclide which may be enclosed in a sealed container made of gold, titanium, stainless steel, or platinum and intended for medical purposes to be placed onto a body surface or into a body cavity or tissue as a source of nuclear radiation for therapy.
- (b) **Classification.** Class II (special controls). A prostate seeding kit intended for use with a radionuclide brachytherapy source only is exempt from the premarket notification procedures in subpart E of part 807 of this chapter subject to the limitations in § 892.9.

[48 FR 53047, Nov. 23, 1983, as amended at 84 FR 71819, Dec. 30, 2019]

§ 892.5740 Radionuclide teletherapy source.

- (a) **Identification.** A radionuclide teletherapy source is a device consisting of a radionuclide enclosed in a sealed container. The device is intended for radiation therapy, with the radiation source located at a distance from the patient's body.
- (b) **Classification.** Class I (general controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter, subject to the limitations in § 892.9.

[53 FR 1567, Jan. 20, 1988, as amended at 59 FR 63015, Dec. 7, 1994; 66 FR 38819, July 25, 2001]

§ 892.5750 Radionuclide radiation therapy system.

- (a) **Identification.** A radionuclide radiation therapy system is a device intended to permit an operator to administer gamma radiation therapy, with the radiation source located at a distance from the patient's body. This generic type of device may include signal analysis and display equipment, patient and equipment supports, treatment planning computer programs, component parts (including beam-limiting devices), and accessories.
- (b) **Classification.** Class II.

§ 892.5770 Powered radiation therapy patient support assembly.

- (a) **Identification.** A powered radiation therapy patient support assembly is an electrically powered adjustable couch intended to support a patient during radiation therapy.
- (b) **Classification.** Class II.

§ 892.5780 Light beam patient position indicator.

- (a) **Identification.** A light beam patient position indicator is a device that projects a beam of light (incoherent light or laser) to determine the alignment of the patient with a radiation beam. The beam of light is intended to be used during radiologic procedures to ensure proper positioning of the patient and to monitor alignment of the radiation beam with the patient's anatomy.
- (b) **Classification.** Class I (general controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter, subject to the limitations in § 892.9.

[53 FR 1567, Jan. 20, 1988, as amended at 61 FR 1125, Jan. 16, 1996; 66 FR 38819, July 25, 2001]

§ 892.5840 Radiation therapy simulation system.

- (a) **Identification.** A radiation therapy simulation system is a fluoroscopic or radiographic x-ray system intended for use in localizing the volume to be exposed during radiation therapy and confirming the position and size of the therapeutic irradiation field produced. This generic type of device may include signal analysis and display equipment, patient and equipment supports, treatment planning computer programs, component parts, and accessories.
- (b) **Classification.** Class II.

§ 892.5900 X-ray radiation therapy system.

- (a) **Identification.** An x-ray radiation therapy system is a device intended to produce and control x-rays used for radiation therapy. This generic type of device may include signal analysis and display equipment, patient and equipment supports, treatment planning computer programs, component parts, and accessories.
- (b) **Classification.** Class II.

§ 892.5930 Therapeutic x-ray tube housing assembly.

- (a) **Identification.** A therapeutic x-ray tube housing assembly is an x-ray generating tube encased in a radiation-shielded housing intended for use in radiation therapy. This generic type of device may include high-voltage and filament transformers or other appropriate components when contained in radiation-shielded housing.
- (b) **Classification.** Class II.

Subpart G—Miscellaneous Devices

§ 892.6500 Personnel protective shield.

- (a) **Identification.** A personnel protective shield is a device intended for medical purposes to protect the patient, the operator, or other persons from unnecessary exposure to radiation during radiologic procedures by providing an attenuating barrier to radiation. This generic type of device may include articles of clothing, furniture, and movable or stationary structures.
- (b) **Classification.** Class I (general controls). The device is exempt from the premarket notification procedures in subpart E of part 807 of this chapter subject to § 892.9.

[53 FR 1567, Jan. 20, 1988, as amended at 61 FR 1125, Jan. 16, 1996; 65 FR 2323, Jan. 14, 2000]

§ 892.6510 Cream for x-ray attenuation.

- (a) **Identification.** A cream for x-ray attenuation is a sterile cream intended for use as a radiation shield. It is intended to be applied to the user's hand before donning gloves, or it may be applied on a glove on the hand, followed by donning a second glove. Cream for x-ray attenuation is intended to be used during medical procedures in which hands are necessarily exposed to radiation to offer some degree of protection from radiation exposure in the diagnostic imaging range of up to 130 kVp. This may include surgical procedures that require the use of fluoroscopy or radiography or other procedures. Cream for x-ray attenuation is not intended to be used in or adjacent to the primary x-ray beam or the transmitted beam and should not be used in lieu of a Radiographic Procedure Glove, which is used in radiography for those studies requiring the physician's hand or forearm be in the direct path of the primary x-ray beam.
- (b) **Classification.** Class II (special controls). The special controls for this device are:
 - (1) Design verification and validation must include documentation of results from safety and effectiveness testing. The results from safety and effectiveness testing must include:
 - (i) Biocompatibility data consistent with the intended use for the device;
 - (ii) Sterilization, packaging, and expiration date testing; and

- (iii) Nonclinical and/or clinical performance testing representative of “as use” conditions demonstrating:
 - (A) Compatibility to the type(s) of surgical glove (e.g., latex, nitrile, vinyl) to be used with the device;
 - (B) Attenuation performance; and
 - (C) Proper application of the device.
- (2) Labeling must include:
 - (i) A statement that the device is sterile and an expiration date.
 - (ii) A boxed warning statement prominently placed in all labeling material for these devices. That boxed warning statement must read: “The device is not intended to be used in or adjacent to the primary X-ray beam or transmitted beam and should not be used in lieu of a Radiographic Procedure Glove, which is used in radiography for those studies requiring the physician's hand or forearm be in the direct path of the primary X-ray beam.”
 - (iii) The methods and results from nonclinical and/or clinical performance testing representative of “as use” conditions demonstrating the amount of attenuation the device provides to the end user at 60, 80, 100, and 120 kVp.
 - (iv) Validated instructions for use for device application and a statement of how often the device must be removed and reapplied for effective shielding.
 - (v) Identification of the type(s) of surgical glove (e.g., latex, nitrile, vinyl) that is compatible for use with the device.

[90 FR 27238, June 26, 2025]